# Zara Sarkar

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## SUMMARY

**AI Engineer (MS Computer Science)** specializing in **Python-based machine learning** and **agentic orchestration**. Proven track record of driving business value by **reducing system latency by 95%**, optimizing **Random Forest classifiers**, and automating intelligence layers that **cut manual testing by 50%**. Expert in **Python**, **scikit-learn**, and **LangChain**, with a focus on building **production-ready NLP pipelines**, **unsupervised clustering**, and **scalable RAG architectures**.

## EDUCATION

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|---|---|
| Computer Science (Master's of Science) | **Northeastern University** |
| Data Analytics (Bachelor's of Science) | **Bentley University** |

## SKILLS

**Technical Skills**

Python, R, Spark, Java, C, SQL, Postgres, AWS Glue, AWS Athena, Databricks, AWS Dynamo

**Relevant Courses**

Data Mining, Machine Learning, Causal Inference, Data Structures and Algorithms, Databases

## EXPERIENCE

AI Engineer (Newark, NJ) — **Prudential Global Investment Management** — Sept 2025 - Present

- Architected a **multi-agent** orchestration layer using **Python and Microsoft AutoGen** to automate meeting intelligence, enabling dynamic schema traversal of **Salesforce** data.
- Optimized system latency by **95% (from 10s to 530ms)** by implementing application-side pre-warmed cache and **Thread Pool Executors** to parallelize queries.
- Engineered an automated **RAG pipeline** using **Langchain** and **Microsoft AutoGen** to create custom validation datasets, reducing manual testing by **50%**.
- Increased system accuracy by **25%** through the implementation of critic agents to autonomously score and validate outputs.
- Presented **Meeting AI** to stakeholders

Data Science Co-op (Newark, NJ) — **Prudential Global Investment Management** — Jul 2025 - Sept 2025

- Developed a chatbot pipeline to translate Natural Language queries into SQL using **LangChain** and **GenAI**.
- Built a **semantic caching layer** using **GloVe** embeddings, **BM25**, and **Postgres vector databases**, reducing LLM API calls by **25%**.
- Benchmarked **Random Forest and Multinomial Regression** classifiers in **scikit-learn** to optimize LLM routing and feature selection.
- Conducted **unsupervised NLP analysis** (**LDA, TF-IDF**) on user logs to identify intent patterns, improving user segmentation clustering by 20%.
- Integrated **AWS DynamoDB and Azure SQL** data pipelines, reducing computation time for downstream analytics by **30%**.

Data Science Intern (Boston, MA) — **Management Solutions** — June 2023 - Aug 2023

- Applied **Logistic and Poisson GLMs via Python statsmodels** to evaluate predictors in climate risk models; presented insights to Director of Data Science

Data Analyst Intern (Woburn, MA) — **American Tower** — May 2022 - Aug 2022

- Designed **materialized views** in **SQL** to consolidate data and increased reporting speed by **50%** using **CTEs and JOINs**.

## PROJECTS

**Luxxify: Realtime Makeup Recommender**

- Designed personalized **RandomForest machine learning recommender (sci-kit learn)** for makeup products, integrating domain-specific embeddings into ensemble pipelines to optimize user preference predictions
- Trained custom **Word2Vec embeddings (gensim)** on **100MB of Reddit makeup reviews** to generate interpretable semantic features, enhancing model understanding of nuanced product preferences
- Developed and optimized feature engineering for NLP-based inputs, improving **RandomForest classifier** performance and **boosting F1 score by 30%**.
- Applied oversampling and undersampling strategies to balance datasets and prevent data leakage, resulting in **12% improvement in model robustness**.
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Implemented scalable processing of text-based features **(BeautifulSoup4, Selenium, Asyncio)**, maintaining performance across **300k+ user reviews**

**Unsupervised Clustering Methods in Python**

- Developed **Poisson mixture model** in Python using **NumPy** and **PyTorch** to fit count data with heterogeneous rates, implementing **Expectation-Maximization (EM)** for parameter estimation and validating model fit through **likelihood-based metrics**

**Data Structures and Algorithms in C**

- Programmed **LinkedLists, Hash Tables, Trees, and Queues** and utilized **algorithms**: backtracking, BFS, DFS, recursion
- Implemented a Unix-style shell in **C** with **memory management (malloc/free)**, **concurrency**, and zombie process handling